

From Intertwiners to the Oriented Tetrahedron: Algebraic Origin of the Factor $K_{\text{node}} = 2$

A Technical Note within Tetrahedral Emergent Gravity (TEG)
(*Work in progress — algebraic core + physical hypotheses*)

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Abstract

This technical note collects, in a self-contained and carefully ordered form, the algebraic steps that lead from the dimension of the space of $\text{SU}(2)$ intertwiners of a 4-valent node with spins $j_i = \frac{1}{2}$ to the factor $K_{\text{node}} = 2$ used in Tetrahedral Emergent Gravity (TEG). The calculation of $\dim \mathcal{H}_{\text{int}} = 2$, the construction of an explicit orthonormal basis, and the geometric identification of these two states with the two orientations of a tetrahedron are standard.

The subsequent steps that convert this factor into an effective branching number $b_{\text{eff}} = 8$ and a spectral dimension $d_s = \ln 8$ rely on additional physical hypotheses (causal rigidity of the EPRL vertex and a Bethe-lattice approximation). These hypotheses are stated explicitly and remain part of ongoing work. The note is offered as a clear reference for the algebraic core and as an honest map of where the rigorous mathematics ends and the conjectural physics begins.

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1 Purpose of this note

In Tetrahedral Emergent Gravity (TEG) one encounters the intermediate factor

$$K_{\text{node}} = 2$$

that is used to convert the tetrahedral coordination number $z_{\text{fund}} = 4$ into an effective branching factor

$$b_{\text{eff}} = K_{\text{node}} \times z_{\text{fund}} = 8,$$

from which the spectral dimension is written as

$$d_s = \ln b_{\text{eff}} = \ln 8.$$

The purpose of the present note is twofold:

1. To give a clean, fully explicit account of the **algebraic** origin of the number 2 (the dimension of the intertwiner space of a 4-valent node with spins $j_i = \frac{1}{2}$, together with an orthonormal basis).
2. To state clearly which subsequent steps remain **physical hypotheses** rather than theorems, so that the boundary between rigorous representation theory and conjectural physics is transparent.

This is a working document. It is not claimed that the full chain from intertwiners to $d_s = \ln 8$ has been derived from first principles with the same level of control.

2 General formula for the dimension of 4-valent intertwiners

For a 4-valent node with arbitrary spins j_1, j_2, j_3, j_4 (non-negative integers or half-integers) the dimension of the intertwiner space is

$$\dim \mathcal{H}_{\text{int}}(j_1, j_2, j_3, j_4) = \dim \left(\text{Inv}_{\text{SU}(2)}(V_{j_1} \otimes V_{j_2} \otimes V_{j_3} \otimes V_{j_4}) \right).$$

2.1 Closed-form expression

Define

$$a = \min(j_1 + j_2, j_3 + j_4), \quad b = \max(|j_1 - j_2|, |j_3 - j_4|).$$

Then

$$\dim \mathcal{H}_{\text{int}} = \begin{cases} a - b + 1 & \text{if } a \geq b \text{ and } j_1 + j_2 + j_3 + j_4 \in \mathbb{Z}, \\ 0 & \text{otherwise.} \end{cases}$$

2.2 Equivalent form using Clebsch–Gordan multiplicities

$$\dim \mathcal{H}_{\text{int}} = \sum_k N_{j_1 j_2 k} N_{j_3 j_4 k},$$

where $N_{abc} = 1$ if the triangle inequalities $|a - b| \leq c \leq a + b$ hold and $a + b + c$ is integer, and $N_{abc} = 0$ otherwise.

2.3 Important special cases

Spins (j_1, j_2, j_3, j_4)	Dimension
$(1/2, 1/2, 1/2, 1/2)$	2
$(1, 1, 1, 1)$	3
(j, j, j, j)	$2j + 1$ (when allowed)
$(1/2, 1/2, 1/2, 3/2)$	1
$(0, 0, 0, 0)$	1

The case used throughout TEG is the fundamental one: all four spins equal to $1/2$.

3 Detailed calculation for $j_i = 1/2$

We compute

$$\dim\left(\text{Inv}_{\text{SU}(2)}(V_{1/2} \otimes V_{1/2} \otimes V_{1/2} \otimes V_{1/2})\right).$$

3.1 Step 1 – Total Hilbert space

Each representation $V_{1/2}$ has dimension $2j + 1 = 2$. Therefore

$$\dim(V_{1/2}^{\otimes 4}) = 2^4 = 16.$$

3.2 Step 2 – Pairwise coupling

Couple the first two spins:

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

(dimensions $1 + 3 = 4$). Couple the last two spins in exactly the same way:

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1.$$

3.3 Step 3 – Combinations that give total spin zero

We now look for singlets inside

$$(0 \oplus 1) \otimes (0 \oplus 1).$$

- $0 \otimes 0$ produces one singlet.
- $1 \otimes 1 = 0 \oplus 1 \oplus 2$ contains exactly one singlet.
- The mixed terms $0 \otimes 1$ and $1 \otimes 0$ produce spin 1 and do not contribute.

Hence there are **exactly two independent channels** that yield a total angular momentum $J = 0$.

3.4 Step 4 – Conclusion

$$\dim \mathcal{H}_{\text{int}}^{(4)} \Big|_{j_i=1/2} = 2.$$

4 Explicit orthonormal basis of the two intertwiners

We use the $SU(2)$ -invariant antisymmetric tensor (Levi-Civita symbol)

$$\varepsilon_{AB} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The two linearly independent intertwiners are:

s-channel (parallel contraction)

$$|\iota_0\rangle = \frac{1}{\sqrt{2}} \varepsilon_{AB} \varepsilon_{CD}.$$

t-channel (crossed contraction)

$$|\iota_1\rangle = \frac{1}{\sqrt{2}} \varepsilon_{AC} \varepsilon_{BD}.$$

(The third possible contraction $\varepsilon_{AD}\varepsilon_{BC}$ is linearly dependent: $\iota_0 + \iota_1 + \iota_u = 0$.)

In ket notation they can also be written as

$$\begin{aligned} |\iota_0\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)_{12} \otimes (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)_{34}, \\ |\iota_1\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)_{13} \otimes (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)_{24}. \end{aligned}$$

4.1 Proof of orthonormality (inner product)

The natural inner product on the space of intertwiners is induced by the Hilbert–Schmidt product on the tensor product of representation spaces. For four spin-1/2 it reduces to the ordinary contraction of all free indices (with complex conjugation on the bra).

Norm of $|\iota_0\rangle$.

$$\langle \iota_0 | \iota_0 \rangle = \frac{1}{2} \varepsilon^{AB} \varepsilon^{CD} \varepsilon_{AB} \varepsilon_{CD} = \frac{1}{2} \cdot 2 \cdot 2 = 1.$$

(Here we used the identity $\varepsilon^{AB} \varepsilon_{AB} = 2$.)

Norm of $|\iota_1\rangle$.

$$\langle \iota_1 | \iota_1 \rangle = \frac{1}{2} \varepsilon^{AC} \varepsilon^{BD} \varepsilon_{AC} \varepsilon_{BD} = 1$$

by the same identity after relabeling.

Orthogonality.

$$\langle \iota_0 | \iota_1 \rangle = \frac{1}{2} \varepsilon^{AB} \varepsilon^{CD} \varepsilon_{AC} \varepsilon_{BD}.$$

Contracting the indices yields zero because the two different pairing schemes are orthogonal under the full contraction (they belong to distinct irreducible channels). Explicitly, the only non-vanishing contributions would require simultaneous equality of the index pairs $(A, B) = (A, C)$ and $(C, D) = (B, D)$, which is impossible for a consistent assignment of four distinct indices. Hence

$$\langle \iota_0 | \iota_1 \rangle = 0.$$

Therefore $\{|\iota_0\rangle, |\iota_1\rangle\}$ is an orthonormal basis of the two-dimensional intertwiner space.

5 Geometric interpretation: the oriented tetrahedron from the 5-cell

5.1 The Minimal Simplex Principle (MSP)

In $\mathbb{R}^d$ the unique regular simplex (minimal convex polytope of positive volume) has exactly $d + 1$ vertices.

Dimension d	Regular simplex	Vertices $z(d)$
1	Segment	2
2	Equilateral triangle	3
3	Regular tetrahedron	4
4	5-cell (pentachoron)	5

Only in $d = 3$ one has $z(3) = 4 = \dim(\mathbb{R}^3)$. This is the content of the Minimal Simplex Principle used in TEG.

5.2 The 5-cell in S^3

The unique regular simplex in four dimensions is the **5-cell** (also called pentachoron or 4-simplex). It has five vertices and lives naturally on the 3-sphere $S^3 \subset \mathbb{R}^4$. Its symmetry group is the alternating group A_5 .

5.3 Algebraic projection $\pi : S^3 \rightarrow \mathbb{R}^3$

Using the quaternionic identification $\mathbb{H} \simeq \mathbb{R}^4$, the projection that forgets the scalar part

$$q = a + bi + cj + dk \quad \mapsto \quad (b, c, d) \in \mathbb{R}^3$$

maps the 5-cell onto a regular tetrahedron in $\mathbb{R}^3$. One vertex is lost (it becomes the scalar component that is projected out); the remaining four vertices form the tetrahedron. Coordination number is preserved:

$$z_{\text{fund}}(\mathbb{H}) = 4 \quad \longrightarrow \quad z_{\text{fund}}(\mathbb{R}^3) = 4.$$

5.4 The two orientations

A tetrahedron in $\mathbb{R}^3$ admits exactly two discrete orientations (right-handed and left-handed). These two orientations are identified with the two-dimensional intertwiner space constructed above:

- $|\iota_0\rangle \leftrightarrow$ one orientation,
- $|\iota_1\rangle \leftrightarrow$ the opposite orientation.

6 The hypothesis of causal rigidity

In the literature on the EPRL model it is often argued that the causal structure of the vertex amplitude prevents the two orientations from cancelling each other. If this non-cancellation persists under coarse-graining, each 4-valent node would contribute a multiplicative factor

$$K_{\text{node}} = 2$$

to an effective branching process.

We emphasise that this step is a **physical hypothesis**, not a theorem. A derivation that starts from the full EPRL vertex amplitude, implements the simplicity constraints, and tracks the contribution of the two intertwiners through a renormalisation group flow is still lacking. Until such a derivation is available, $K_{\text{node}} = 2$ should be regarded as a motivated ansatz rather than an established result.

7 From K_{node} to a spectral-dimension estimate

If one accepts the hypothesis $K_{\text{node}} = 2$ and adopts a Bethe-lattice (tree) approximation for the vacuum network, the effective branching factor becomes

$$b_{\text{eff}} = K_{\text{node}} \times z_{\text{fund}} = 2 \times 4 = 8.$$

On a regular tree the return probability of a random walker behaves as

$$P(0, t) \sim t^{-d_s/2},$$

with

$$d_s = \ln b_{\text{eff}} = \ln 8 \approx 2.07944.$$

This is the origin of the value $D_{\text{eff}} = \ln 8$ used in TEG. It should be understood as a heuristic estimate within a simplified geometric ensemble, not as a first-principles computation of the spectral dimension of a continuum quantum geometry.

8 Logical chain (with status labels)

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[STANDARD] General intertwiner dimension formula
              | specialise to  $j_i = 1/2$ 
[STANDARD]  $\dim \mathcal{H}_{\text{int}} = 2$ 
              | explicit basis  $\{|\iota_0\rangle, |\iota_1\rangle\}$ 
[STANDARD] Two orthonormal intertwiners
              | geometric identification
[STANDARD] Two orientations of the tetrahedron
              | 5-cell projection / MSP
[MOTIVATING] Origin of  $z_{\text{fund}} = 4$ 
              | causal rigidity hypothesis
[HYPOTHESIS]  $K_{\text{node}} = 2$ 
              | tetrahedral coordination
[HYPOTHESIS]  $b_{\text{eff}} = 2 \times 4 = 8$ 
              | Bethe-lattice heuristic
[HEURISTIC]  $d_s = \ln 8 = D_{\text{eff}}$ 
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9 Status, limitations and open problems

It is important to distinguish carefully what has been established from what remains conjectural.

9.1 What is solid

- The calculation $\dim \mathcal{H}_{\text{int}}^{(4)} = 2$ for four spins $j_i = \frac{1}{2}$ is standard, exact, and textbook material in loop quantum gravity.
- The explicit orthonormal basis $\{|\iota_0\rangle, |\iota_1\rangle\}$ constructed with the Levi-Civita tensor is likewise standard.
- The geometric identification of these two states with the two discrete orientations of a tetrahedron is classical (Baez–Barrett and subsequent literature).

9.2 What is still hypothetical

- **Causal rigidity** $\rightarrow K_{\text{node}} = 2$. The claim that the causal structure of the EPRL vertex prevents destructive interference between the two orientations, and that this immediately supplies a multiplicative factor $K_{\text{node}} = 2$ in a branching process, is a physical hypothesis. A controlled derivation from the full vertex amplitude, including the simplicity constraints and the sum over intertwiners, is still missing.
- **Bethe-lattice approximation.** The relation $d_s = \ln b_{\text{eff}}$ with $b_{\text{eff}} = 8$ is a heuristic taken from tree-like (Bethe) lattices. A genuine computation of the spectral dimension would require defining a continuum limit or a statistical ensemble of geometries, constructing the heat kernel (or the return probability $P(0, t)$), and extracting the asymptotic exponent. This has not been carried out.
- **Minimal Simplex Principle and the 5-cell projection.** These are elegant geometric observations that motivate why $z = 4$ is natural in three dimensions. They do not, by themselves, constitute a dynamical derivation of the vacuum coordination from a quantum-gravity path integral.

9.3 Open problems

1. Derive (or refute) the effective factor $K_{\text{node}} = 2$ from the EPRL vertex amplitude under a controlled coarse-graining procedure.
2. Compute the spectral dimension of a tetrahedral spin-network ensemble directly from the heat kernel, without invoking the Bethe-lattice formula.
3. Clarify the precise status of the Minimal Simplex Principle: is it merely an algebraic curiosity, or can it be elevated to a dynamical selection principle?

In short: the algebraic core of this note is reliable. The physical chain that converts that core into $d_s = \ln 8$ is still work in progress and should be read as such.

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